HUNDRED DAYS OF CODE : DAY - 8

NAME - ARVIND KUMAR SAP ID - 590042201

1.Problem Statement - 1

Write a C program to find the cost of making a conical tent when radius, height, and cloth cost per square meter are given.

1.1 Inputs, Outputs & Formula Used

The Formula used for this problem is : pi*r*l

where, pi = 3.14

r = radius, h = height

l = slaint height sqrt(h^(2))+(r^(2))

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.	r:3 h:4 c:10	471	471	PASS
2.	r:5 h:12 c:5	1020.50	1020.50	PASS
3.	r:8 h:20 c:600	324660.81	324660.81	PASS

1.2 The algorithm

1. Start.
2. Declare variables for radius, height, cost_per_unit, slant_height, area, and total_cost.
3. Define the constant pi = 3.14.
4. Read inputs for radius, height, and cost_per_unit.
5. Calculate l = sqrt{r^2 + h^2}.
6. Calculate area = pi * r *l.
7. Calculate total_cost = area cost_per_unit.
8. Print the total_cost.
9. Stop.

1.3 The C Program

```
#include<stdio.h>
#include<math.h>

int main()
{
 float pi, r ,h ,l, area, cost, total_cost;

 printf("Enter the radius: \n");
 scanf("%f",&r);
 printf("Enter the height: \n");
 scanf("%f",&h);
 printf("Enter the cost per meter square: \n");
 scanf("%f",&cost);

 pi = 3.14;
 l = sqrt(r*r+h*h);
 area = pi*r*l;

 total_cost = area*cost;

 printf("The cost for making the tent is :
%.2f\n",total_cost);
 return 0;
}
```

1.4 Output

```
┌──(kali㉿kali)-[~/codes/day_8]
└─$ gcc que1.c -o que1 -lm

┌──(kali㉿kali)-[~/codes/day_8]
└─$ ./que1
Enter the radius:
3
Enter the height:
4
Enter the cost per meter square:
10
The cost for making the tent is : 471.00

┌──(kali㉿kali)-[~/codes/day_8]
└─$ ./que1
Enter the radius:
5
Enter the height:
12
Enter the cost per meter square:
5
The cost for making the tent is : 1020.50

┌──(kali㉿kali)-[~/codes/day_8]
└─$ ./que1
Enter the radius:
8
Enter the height:
20
Enter the cost per meter square:
600
The cost for making the tent is : 324660.81
```

2.Problem Statement - 11

Write a C program to find the area swept by a minute hand of a clock when its length and time in minutes are given.

2.1 Inputs, Outputs & Formula Used

The Formulas used for this problem is : 1. (t/60)*pi*r*r

where t = time in minutes
pi = 3.14
r = length of the minute hand

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.	t:15, r:7	38.47	38.47	PASS
2.	t:10, r:5	13.08	13.08	PASS
3.	t:100000000, r:22222	2584310544 596992.00	2584310544 596992.00	PASS

2.2 The algorithm

1. Start.
2. Declare variables for length, time, and area.
3. Define the constant pi = 3.14.
4. Read inputs for length and time.
5. Calculate area = (time / 60.0) *pi *length *length$.
6. Print calculated area.
7. Stop.

2.3 The C Program

```
#include<stdio.h>

int main()
{
 float pi, r, area;
 int time;

 printf("Enter the time swept by the minute hand (mins) :
\n");
 scanf("%d",&time);
 printf("Enter the length of the minute hand : \n");
 scanf("%f",&r);

 pi = 3.14;

 area = (time/60.0)*pi*r*r;

 printf("The area swept by the minute hand is : %.2f\n",area);
 return 0;
}
```

2.4 Output

```
┌──(kali㉿kali)-[~/codes/day_8]
└─$ gcc que2.c -o que2

┌──(kali㉿kali)-[~/codes/day_8]
└─$ ./que2
Enter the time swept by the minute hand (mins) :
15
Enter the length of the minute hand :
7
The area swept by the minute hand is : 38.47

┌──(kali㉿kali)-[~/codes/day_8]
└─$ ./que2
Enter the time swept by the minute hand (mins) :
10
Enter the length of the minute hand :
5
The area swept by the minute hand is : 13.08

┌──(kali㉿kali)-[~/codes/day_8]
└─$ ./que2
Enter the time swept by the minute hand (mins) :
100000000
Enter the length of the minute hand :
22222
The area swept by the minute hand is : 2584310544596992.00

┌──(kali㉿kali)-[~/codes/day_8]
└─$ 
```